A Jaynes-Cummings model of a black hole: effective semiclassicality, branch-level semiclassicality, and unitarity

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Abstract

A Jaynes-Cummings model of a black hole serves as a proof-of-concept for the black hole information problem. We distinguish effective semiclassicality, the classical behavior of macroscopic observables guaranteed by the central limit theorem, from branch-level semiclassicality, the assumption that EFT entanglement structure holds uniformly across all branches in a quantum superposition. Simulations and an exact analytic solution demonstrate unitarity while effective semiclassicality is maintained, but only because the empty branch forces a departure from branch-level semiclassicality at late times. The entropy decrease arises entirely from the growing weight of the empty-branch component. The assumption underlying the small corrections theorem, that branch-level EFT holds throughout evaporation, is not a necessary condition for consistency. A conceptual extension of dual semiclassicality to replica wormholes in double-scaled JT gravity is made, and separating the two notions reshapes how the information paradox is framed.

Keywords: Jaynes-Cummings model, black hole information problem, semiclassical physics, qubit model, dual semiclassicality

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CONTENTS

I. INTRODUCTION

A central conceptual tension in the black hole information debate concerns what semi-classicality actually means, and how much of it must be sacrificed to restore unitarity [1, 2]. *Conventional* treatments of contemporary quantum gravity approaches, notably replica wormholes and the islands formula [3, 4], recover a Page curve by allowing the effective description to depart from semiclassical expectations at late times, bypassing the small corrections theorem [5]. The departures happen despite these frameworks being initially guided by the semiclassical framework. Since we are yet to observe semiclassicality breaking down macroscopically, the reasons why this departure is admissible have been debated. Can the derivations from the *quasi-semiclassical* framework be sufficient to reverse late-time semiclassical conclusions? Do these departures predict significant drama, firewalls or change in the nature of gravity around the horizon [6–8] or can we maintain no drama? The natural or naive interpretation of islands suggests that Hilbert space factorization between the black hole interior and the exterior breaks down - is it really the case? Complications arise due to the fact that many of the approaches with late-time semiclassicality departures are cast within holographic AdS/CFT [9], and it remains to be seen what are artifacts of AdS/CFT and what are general.

This paper demonstrates, via a Jaynes-Cummings (JCM) toy model, that the apparent tension can more naturally be dissolved once two distinct notions of semiclassicality - effective semiclassicality and branch-level semiclassicality - are carefully separated (see their definitions in Section II H). This requires no Hilbert space factorization breakdown between the black hole interior and the exterior and a single EFT is maintained within the JCM model, though the JCM toy model can be used to analyze how and why Hilbert space factorization may break down, as well as why different EFTs may emerge despite effective semiclassicality being maintained (see Section II F). As we will see in Section III, this does not conflict with the existing approaches, only requiring recast. Section III also confirms that the JCM model intuitions are not finite-dimensional artifacts.

We stress that the paper does *not* claim the small corrections theorem is wrong. The theorem is valid given its assumptions. What the JCM model shows is that the assumption of uniform branch-level EFT entanglement structure is not necessary for consistency, and that relaxing it, in the precise way dictated by correct treatment of the 'empty black hole branch', is sufficient to restore unitarity while preserving effective semiclassicality. Furthermore, the intuition of the 'empty black hole branch' can be generalized, as seen in Section III for the replica wormholes case, with the basic insights remaining the same.

In general, any unitary model of black hole evaporation satisfying (i) energy conservation, (ii) a unique vacuum ground state for the black hole (the no-hair condition for a given charge and angular momentum), and (iii) quantum superposition of branches, will necessarily develop something resembling an empty-branch component with nonzero amplitude, and the correct EFT treatment of that branch will force the entanglement entropy to turn over. The JCM model provides a clean, analytically tractable proof of concept for this general principle, in the same spirit as AdS/CFT toy models that elucidate gravitational concepts without claiming to describe the entirety of realistic quantum gravity.

Throughout the paper we assume $\hbar = 1$.

A. Additional historical backgrounds

1. *Wen's JCM model of a black hole*

A use of JCM for modelling a black hole is not completely new. Reference [10] uses the JCM model with a reflective cavity such that reflected photons can re-interact with a black hole atom, using atoms as black holes rather than a cavity. That setup replicates the Parikh-Wilczek non-thermal spectrum [11], but its periodic entropy curves of incomplete decrease imply it is not a model of full evaporation. The present model differs by placing the black hole as the cavity, with qubits representing outgoing Hawking radiation that fly away permanently, the setup natural for studying complete evaporation.

2. *Small corrections theorem*

We briefly describe the toy qubit model logic of the small corrections theorem [5]. At each step, a single qubit pair is produced around the black hole horizon, one qubit falling in and one radiated away:

$$|\psi^{eff}_{pair,n}\rangle = \cos(\phi)|0^{eff}_{R,n}\rangle|0^{eff}_{BH,b}\rangle + \sin(\phi)|1^{eff}_{R,n}\rangle|1^{eff}_{BH,n}\rangle + O(\epsilon), \quad \epsilon \ll 1. \tag{1}$$

Invoking strong subadditivity, the theorem concludes that $S(R_{\leq n+1}) \geq S(R_{\leq n})+S(BH^{eff}_{n+1})-O(\epsilon)$, so that radiation entropy increases monotonically as long as EFT entanglement persists. With the JCM model we show that, without violating EFT itself, the branch-level assumption

$$S(BH^{eff}_{n+1}) = -\cos^2\phi\,\ln\cos^2\phi - \sin^2\phi\,\ln\sin^2\phi \tag{2}$$

fails at late times (in particular for the 'empty black hole' branch), thereby invalidating the premise from which monotone increase is derived. The theorem's logic is sound; its assumption is what breaks down.

II. A JCM MODEL OF A BLACK HOLE

A. Setup

Each kth qubit (atom in the original JCM), initially in $|0\rangle$, interacts sequentially with the black hole (optical cavity in the original JCM) for duration Δt_k from $t = \sum_{j=1}^{k-1} \Delta t_j$ to $t = \sum_{j=1}^{k} \Delta t_j$, then flies away, never interacting with the black hole again. The initial black hole state is $|N\rangle$ at $t = 0$, with N the particle number; there are M qubits in total.

The free Hamiltonians and JCM interaction are:

$$H_{\text{rad}} = \omega_a \sigma_+ \sigma_- = \omega_a |1\rangle\langle 1|, \quad H_{BH} = \omega_a a^\dagger a, \quad H_I = g(\sigma_+ a + \sigma_- a^\dagger), \tag{3}$$

with $[a, a^\dagger] = 1$, with $\sigma_+ = |1\rangle\langle 0|$ and $\sigma_- = |0\rangle\langle 1|$. The interaction conserves total particle number and non-interaction energy; ω_a is irrelevant to the entropy analysis.

B. Variable interaction time

The unitary for the kth interaction is:

$$U_k |0, n\rangle = \cos(g\sqrt{n}\,\Delta t_k)|0, n\rangle - i \sin(g\sqrt{n}\,\Delta t_k)|1, n-1\rangle. \tag{4}$$

Choosing $\Delta t_k = \pi/(4g\sqrt{\langle n_k\rangle})$ reproduces the small corrections theorem's semiclassical condition (up to phase) for $\langle n_k\rangle \in \mathbb{N}$.

C. Fixed interaction time

Setting $\Delta t_k = \Delta t = \pi/(4g\sqrt{N})$ reproduces the semiclassical condition only for $n = N$.

D. Simplified model

Modifying the fixed-time unitary to:

$$\begin{gathered} U|0,n\rangle = \tfrac{1}{\sqrt{2}}\big(|0,n\rangle + |1,n-1\rangle\big), \quad n>0, \\ U|0,0\rangle = |0,0\rangle, \end{gathered} \tag{5}$$

conserves non-interaction energy and is analytically tractable. The conclusions are qualitatively identical across all model variants. We provide the exact analytic solution of the simplified model for the reduced black hole density matrix in the appendix.

E. Effective semiclassicality via the central limit theorem

Reducing the interaction time by factor k_r (or equivalently reducing the coupling g) so that more qubits M interact within the same physical time, the simplified model gives:

$$U|0,n\rangle = \cos\!\big(\pi/(4k_r)\big)|0,n\rangle + \sin\!\big(\pi/(4k_r)\big)|1,n-1\rangle, \quad n>0. \tag{6}$$

Ignoring the empty branch (valid for early times), the variance per qubit is:

$$\mathrm{Var}[E_{k,\mathrm{rad}}] = \sin^2\!\big(\tfrac{\pi}{4k_r}\big) - \sin^4\!\big(\tfrac{\pi}{4k_r}\big) \approx \frac{\pi^2}{16k_r^2} \quad \text{for large } k_r. \tag{7}$$

By the CLT, the total emitted energy over K qubits satisfies:

$$E_{\mathrm{rad},K} - K\langle E_{k,\mathrm{rad}}\rangle \xrightarrow{d} \mathcal{N}(0, K\sigma^2). \tag{8}$$

For large k_r the fractional variance $K\sigma^2/(K\langle E\rangle)^2 \to 0$, so total emitted energy converges to its semiclassical prediction. Slicing time into sub-intervals and applying CLT to each, effective semiclassicality is established throughout the evaporation. This is precisely the sense in which semiclassicality is maintained: macroscopic energy observables behave classically. It does *not* imply that the quantum state, branch-by-branch, is well described by one EFT on the same background.

There is no radiation in case of the 'empty black hole branch' and therefore its variance is zero. For a late-time superposition, this is reflected in smaller $\langle E\rangle$ and its variance per sub-interval energy observable. Therefore, effective semiclassicality remains undisturbed - in fact, from the macroscopic observable point of view, late times are more classical, and there is no high-energy drama.

F. On the empty black hole branch and the unique vacuum

The physical motivation for the entropy turnaround rests on a simple but important observation about EFT near a black hole horizon.

EFT on a non-empty Schwarzschild background predicts lasting entanglement (Hawking pairs) at asymptotic infinity, with radiation temperature increasing as the black hole loses mass. In contrast, EFT on Minkowski spacetime predicts no lasting entanglement. These predictions are *incompatible*: the metric limit of Schwarzschild spacetime as the black hole empties is Minkowski spacetime, yet the Hawking temperature in this limit diverges. The two limits produce different entanglement stories. Since the Minkowski limit governs the actual physics of the empty branch (if the metric limit is to be trusted), EFT must treat the empty branch differently: it does not produce lasting horizon entanglement.

In terms of an intuition within the islands proposal: some information on the black hole interior is non-existent and already evaporated away, and the entanglement wedge of radiations has to include parts of the black hole interior. This is more precisely represented by a superposition with some branches having all information evaporated away already. This is bound to happen as long as gravity is quantized.

Now consider the unique ground state. Dividing the universe Hamiltonian as $H = H_{\text{ext}} + H_{BH} + H_I$, suppose the exterior remains near its ground state around the horizon (any excitation radiates away). The black hole must then emit energy until it reaches $|0\rangle$. By the no-hair principle, with no charge or angular momentum, the ground state $|0\rangle$ is unique, carrying a pure state for the black hole interior. Therefore, on any branch that has reached complete evaporation, the black hole interior is pure, and the exterior is consequently purified.

For semiclassical gravity, this correction cannot easily be incorporated: the semiclassical interaction Hamiltonian H_I treats all branches uniformly and cannot account for the non-generation of entanglement on the empty branch. For quantized gravity, branches are distinguishable, and the empty branch contributes correctly to the total quantum state. In relation to Section III, we could say entanglement islands and replica wormholes are all about incorporating these corrections without abandoning quasi-semiclassical frameworks.

The 'empty black hole branch' (or equivalently, the branch where information is completely depleted from the black hole) also suggests the conceptual intuition as to why exterior radiations can have entanglement wedges that are both in the black hole exterior and the interior ('islands'): there may not be a black hole interior subspace inside the Hilbert space of the universe for the branch resembling 'the empty black hole branch', and semiclassical Hilbert space factorization between the interior and the exterior breaks down globally, creating an illusion of drama. This breakdown is only an artifact of a semiclassical approach, and for each branch, effective field theory is not actually broken - just that the empty branch requires a different EFT treatment, and incorporating this back to the usual EFT for non-empty branches gives an illusion of large corrections.

This argument, like the JCM toy model itself, should be understood as a proof of concept. It assumes the standard EFT and no-hair expectations; if either fails (e.g., fuzzballs or other horizon structures), the argument requires modification.

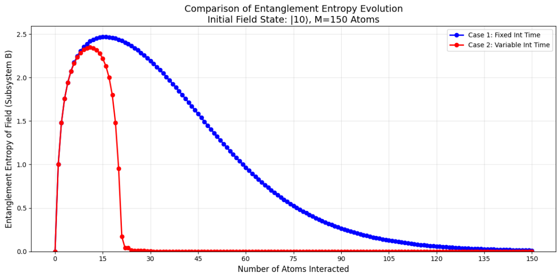
G. Simulation results and analysis

Simulation results (Fig. 1) confirm unitarity after complete evaporation, across all model variants and parameter choices. The Page time $M^* \approx 2N$ (in terms of the number of qubits with completed interactions) predicted by Theorem 2 in the appendix is in good agreement with the observed turnaround. Since interaction time reduction (large k_r) does not change the unitarity conclusion, effective semiclassicality is maintained by Section II E.

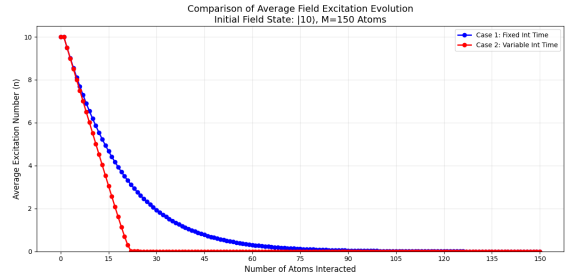
1. Remarks

Theorem 2 in the appendix gives the analytic explanation for the full JCM model as well for the following qualitative explanation: the $|0,0\rangle$ component of the black hole state becomes non-negligible once the expected number of qubit interactions exceeds $\sim 2N$, and the absorbing boundary forces the entropy to decrease. Even when different $|n\rangle$ states are treated as physically distinguishable, the entropy must eventually decrease once the black hole reaches its ground state; nothing is wrong with EFT on the non-empty branches, and the issue is solely that branch-level semiclassicality, applying the same EFT entanglement formula uniformly to *all* branches including the empty one, is incompatible with unitarity.

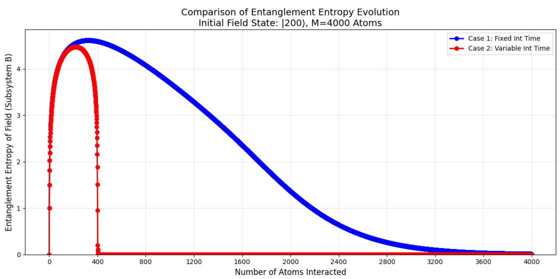
Since this is a toy model, time is indexed by the number of interacting qubits and can be rescaled to match the Page curve slope; the entropy curve's slope is not itself a physical



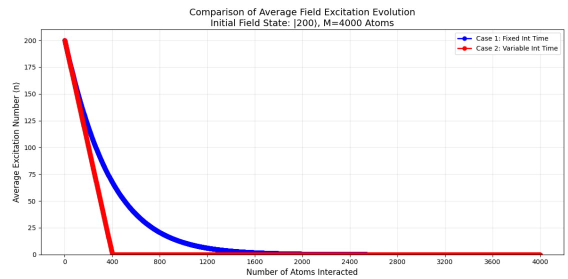
(a) BH entanglement entropy. ($N = 10$, $M = 150$, $g = 1$, $k_r = 1$, full JCM)



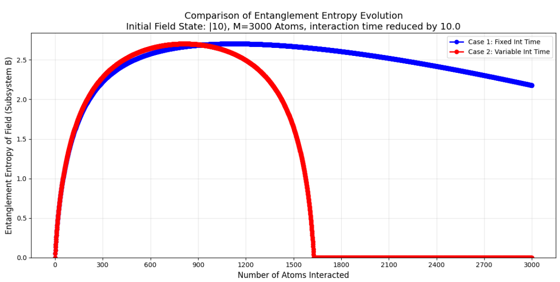
(b) Average BH excitation $\langle n \rangle$. ($N = 10$, $M = 150$, $g = 1$, $k_r = 1$, full JCM)



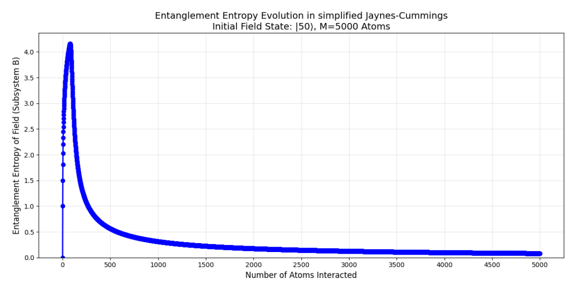
(c) BH entanglement entropy. ($N = 200$, $M = 4000$, $g = 1$, $k_r = 1$, full JCM)



(d) Average BH excitation $\langle n \rangle$. ($N = 200$, $M = 4000$, $g = 1$, $k_r = 1$, full JCM)



(e) BH entanglement entropy. ($N = 10$, $M = 3000$, $g = 1$, $k_r = 10$, full JCM)



(f) BH entanglement entropy. ($N = 50$, $M = 5000$, $g = 1$, $k_r = 1$, simplified model)

FIG. 1: Model simulations. Red/symmetric curves: variable interaction time case; blue curves: fixed interaction time. Page time $M^* \approx 2N$ from Theorem 2 is consistent with the observed turnaround across all cases.

prediction of the model.

Finally, every $|n\rangle$ for $n > 0$ produces the same entanglement behavior, and only $n = 0$ is special. The no-hair feature is already encoded in the $|n\rangle$ structure - $|n\rangle$ is $|n\rangle$ regardless of how it was reached.

H. Effective semiclassicality, branch-level semiclassicality and dual semiclassicality

We now state formally the central conceptual distinction underlying the JCM model.

Definition 1 (Effective semiclassicality)**.** An evaporation model exhibits *effective semiclassicality* if all macroscopic observables (total emitted energy, radiation profile, energy variance) converge to their semiclassical values in the limit of many small interactions (large k_r, large M).

Definition 2 (Branch-level semiclassicality)**.** An evaporation model exhibits *branch-level semiclassicality* if the EFT entanglement structure of Eq. (1) holds on *every* quantum branch of the superposition, including the empty black hole branch, throughout the evaporation.

Analytically, Theorem 2 in the appendix and Section II E together establish:

Theorem 1 (Dual semiclassicality)**.** The simplified JCM model (and the full JCM model qualitatively) exhibits effective semiclassicality throughout, while branch-level semiclassicality necessarily fails at late times. These are compatible: the failure of branch-level semiclassicality is invisible to macroscopic observers (the empty branch contributes zero energy to any macroscopic observable) yet is decisive for the entanglement entropy.

III. THE SCHRÖDINGER-HEISENBERG DUALITY OF ENSEMBLE RANDOMNESS: RECASTING JT GRAVITY REPLICA WORMHOLES

The ensemble randomness over theories invoked in JT gravity [12] can be recast as randomness in the orientation of the late-time scrambled sector within a single fixed theory, in analogy with the equivalence of the Schrödinger and Heisenberg pictures [4]. Here we make this analogy explicit and show how it dovetails with the empty-branch mechanism of Section II F.

A. Two pictures of the same randomness

In double-scaled JT gravity, the gravitational path integral is dual to an average over an ensemble of random matrix Hamiltonians $H(\alpha)$ under matrix potential $V(H)$ defining the probability measure as well as the Schwarzian profile, with α labeling the draw. Replica wormhole geometries connecting different boundary replicas compute genuine ensemble averages such as $\overline{\operatorname{Tr}\rho_S(\alpha)^n}$, and it is precisely the connected (wormhole) contributions to this average that restore a unitary Page curve [3, 4].

This "ensemble of theories" picture sits uneasily with the fact that an observer inhabits a single universe governed, presumably, by one fixed Hamiltonian. An equivalent "subspace" picture removes this tension (see also [4], which implicitly suggests such a formalism): fix the total Hamiltonian $H = H_{\text{base}} + H_{\text{post}} + H_{\text{int}}$ once and for all, and let α instead label which N-dimensional, Haar-random subspace $\mathcal{V}_\alpha \subset \mathcal{H}$ the late-time scrambled sector occupies. The instance-dependent quantity is no longer the Hamiltonian but the embedding $\Phi_\alpha : \mathcal{V}_\alpha \hookrightarrow \mathcal{H}$. Schematically (ignoring H_{base} contributions for both sides without loss of generality),

$$\overline{\operatorname{Tr} e^{-iH(\alpha)t}}\bigg|_{\substack{\text{ensemble of}\\ \text{Hamiltonians}}} \quad \longleftrightarrow \quad \overline{\operatorname{Tr} e^{-i\,\Phi_\alpha^\dagger H_{\text{post}} \Phi_\alpha\, t}}\bigg|_{\substack{H_{\text{post}}\ \text{fixed,}\\ \Phi_\alpha\ \text{Haar-random}}}, \tag{9}$$

Both sides reproduce the same Wigner-Dyson spectral statistics, and hence the same SFF dip-ramp-plateau and Page-averaged entropy, since they agree on expectations (expected value). The two sides of Eq. (9) are not competing hypotheses; they are two bookkeepings of the same Haar randomness predictions, exactly as the Schrödinger and Heisenberg pictures move time dependence between states and operators without changing any expectation value. The ensemble side is the more natural "stepping stone" for computing averaged spectral diagnostics; the subspace side is the more natural stepping stone for asking what happens to a single universe instance.

This subspace reorganization has a direct relation to the empty-branch mechanism of Section II F. There, the absorbing boundary at $n = 0$, the unique no-hair ground state into which the black hole sector is eventually funneled, is what generates the growing weight $P_0^{(M)}$ on the purified branch. In the language of Eq. (9), this absorbing structure plays the role of H_{post} together with the Haar-random embedding Φ_α: just as $\Phi_\alpha^\dagger H_{\text{post}} \Phi_\alpha$ governs the late-time spectral statistics responsible for the ramp and plateau, and hence for purification, in the SFF, the late-time dynamics that drive $P_0^{(M)} \to 1$ are precisely the dynamics that funnel the JCM black hole sector into its unique vacuum. The empty-branch correction is therefore not an addition external to the Schrödinger-Heisenberg-type duality of Eq. (9); it is the finite-dimensional, fully explicit instance of the same late-time, no-hair-driven reorganization that H_{post} and Φ_α implement more abstractly in double-scaled JT gravity.

IV. CONCLUSION

The distinction between effective semiclassicality (macroscopic observables are classical) and branch-level semiclassicality (EFT entanglement structure holds on every quantum branch) clarifies what must and what need not be sacrificed to restore unitarity in black hole evaporation.

The JCM model demonstrates, via simulation and exact analytic solution, that a unitary Page curve is achievable while preserving effective semiclassicality. The entropy turnaround occurs around $M^* \approx 2N$ (Page time), and Theorem 2 shows analytically that this turnaround is caused entirely by the growing weight of the empty black hole branch, a contribution that is macroscopically invisible but quantum-mechanically decisive.

The assumption underlying the small corrections theorem, that EFT entanglement structure holds branch-by-branch throughout evaporation, is shown to be incompatible with unitarity. The theorem itself is valid; its assumption is what fails, and fails in a way that is consistent with effective semiclassicality.

Three conditions sufficient for this dual semiclassicality are identified: energy conservation, unique vacuum (no-hair), and quantum (gravitational) superposition. Any unitary model satisfying these conditions will exhibit the same entropy turnaround. The JCM model is a proof of concept for this general principle, analogous in spirit to AdS/CFT toy models that elucidate gravitational concepts without being literal descriptions of quantum gravity.

EFT is safe, effective semiclassicality is preserved, branch-level semiclassicality breaks down in the only way consistent with unitarity, and this breakdown is macroscopically unobservable.

DATA AVAILABILITY AND DECLARATION OF INTERESTS

The full JCM model code is available at `https://mkimacad.github.io/bh_jaynes_cummings/codes/jaynes_improved.py` and the simplified model code at `https://mkimacad.github.io/bh_jaynes_cummings/codes/jaynes_simplified.py`. The author(s) have no

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Appendix A: Exact analytic solution of the simplified model

The simplified model admits a closed-form solution for the reduced black hole density matrix.

Lemma 1 (Exact BH state)**.** Starting from $|N\rangle$, after M sequential qubit interactions in the simplified model, the reduced density matrix of the black hole is diagonal:

$$\rho_{BH}^{(M)} = P_0^{(M)}|0\rangle\langle 0| + \sum_{n=1}^{N} P_n^{(M)}|n\rangle\langle n|, \tag{A1}$$

where

$$P_n^{(M)} = \binom{M}{N-n} 2^{-M}, \quad 1 \le n \le N, \quad n \ge N - M, \tag{A2}$$

$$P_0^{(M)} = 1 - \sum_{j=0}^{\min(N-1,M)} \binom{M}{j} 2^{-M} = P\big(\mathrm{Bin}(M, \tfrac{1}{2}) \ge N\big). \tag{A3}$$

Proof. The interaction operator, applied to a qubit in $|0\rangle$ and the black hole in $|n\rangle$, followed by tracing over the outgoing qubit, yields the Kraus operators:

$$K_0 = \tfrac{1}{\sqrt{2}} \sum_{n>0} |n\rangle\langle n| + |0\rangle\langle 0|, \quad K_1 = \tfrac{1}{\sqrt{2}} \sum_{n>0} |n-1\rangle\langle n|. \tag{A4}$$

Acting on a diagonal state $\rho = \sum_n P_n |n\rangle\langle n|$:

$$\rho^{(k+1)} = K_0 \rho^{(k)} K_0^\dagger + K_1 \rho^{(k)} K_1^\dagger \tag{A5}$$

yields the recursion for diagonal entries:

$$P_n^{(k+1)} = \tfrac{1}{2}\big(P_n^{(k)} + P_{n+1}^{(k)}\big), \quad n \ge 1, \tag{A6}$$

$$P_0^{(k+1)} = P_0^{(k)} + \tfrac{1}{2} P_1^{(k)}. \tag{A7}$$

(Note $P_{N+1}^{(k)} = 0$, so $P_N^{(k+1)} = P_N^{(k)}/2$.) Since the walk can only stay or decrease, any path to state $n \ge 1$ after M steps consists of exactly $N - n$ "decreases" out of M steps. A path reaching $n \ge 1$ has never touched the boundary at 0 (since the walk is monotone-decreasing), so the absorbing boundary does not affect these states. The number of such paths is $\binom{M}{N-n}$, each with weight 2^{-M}, giving Eq. (A2). Eq. (A3) follows by conservation of probability. $\square$

Remark 1. The diagonal structure is preserved under the recursion since K_0 and K_1 map $|n\rangle$ to $|n\rangle$ and $|n-1\rangle$ respectively, generating no off-diagonal coherences in the $\{|n\rangle\}$ basis.

From Lemma 1, the von Neumann entropy admits a clean decomposition:

$$S^{(M)} = H_b\big(P_0^{(M)}\big) + \big(1 - P_0^{(M)}\big) S_{\geq 1}^{(M)}, \tag{A8}$$

where $H_b(p) = -p \ln p - (1-p)\ln(1-p)$ is the binary entropy function and $S_{\geq 1}^{(M)}$ is the conditional entropy of the black hole among the non-empty branches ($n \geq 1$).

The behavior of each term is determined by $P_0^{(M)}$:

- $P_0^{(M)} = P(\mathrm{Bin}(M, \frac{1}{2}) \geq N)$ grows monotonically in M from 0 to 1.
- At $M = M^* \approx 2N$ (the *Page time* of the model), $P_0^{(M^*)} \approx \frac{1}{2}$ by the symmetry of the binomial distribution ($P(\mathrm{Bin}(2N, \frac{1}{2}) \geq N) = \frac{1}{2} + \frac{1}{2}\binom{2N}{N} 2^{-2N} \to \frac{1}{2}$ for large N). For $M > 2N$, $P_0^{(M)} > \frac{1}{2}$ and grows toward 1.
- Since $P_0^{(M)} \to 1$ as $M \to \infty$: $H_b(P_0) \to 0$, $(1 - P_0^{(M)}) S_{\geq 1}^{(M)} \to 0$, and hence $S^{(M)} \to 0$.

Theorem 2 (Entropy turnaround from empty branch)**.** In the simplified model: (i) $S^{(M)} \to 0$ as $M \to \infty$ (unitarity); (ii) $S^{(M)}$ achieves its maximum at $M^* \approx 2N$ (Page time); (iii) the entropy decrease is caused entirely by the growth of $P_0^{(M)}$, the weight on the empty black hole branch.

In particular, if $P_0^{(M)} \equiv 0$ is imposed throughout (the branch-level semiclassicality assumption of the small corrections theorem), the renormalized conditional distribution $P_n^{(M)}/(1 - P_0^{(M)})$ spreads over an increasing range of n-values as $M \to \infty$, and $S^{(M)}$ increases without bound. This establishes analytically that the entropy turnaround is not a finite-dimensional artefact but a direct consequence of correctly including the empty branch.

Proof sketch. Parts (i) and (ii) follow from the properties of $P_0^{(M)}$ stated above. For part (iii): with $P_0 = 0$ excluded, the remaining probabilities $P_n^{(M)} = \binom{M}{N-n} 2^{-M}$ for $n \geq 1$ form a truncated binomial distribution whose variance grows as $M/4$, spreading over $O(\sqrt{M})$ states near $n \approx N - M/2$. The entropy of a binomial distribution on $O(\sqrt{M})$ states grows as $\frac{1}{2}\ln M$, confirming unbounded increase. Including P_0 exactly reverses this: once $P_0 > \frac{1}{2}$, the total entropy $S^{(M)}$ is dominated by $H_b(P_0)$ which decreases toward zero. $\square$